

III (Uniform & Exponential Distributions)

Discrete Uniform distribution:

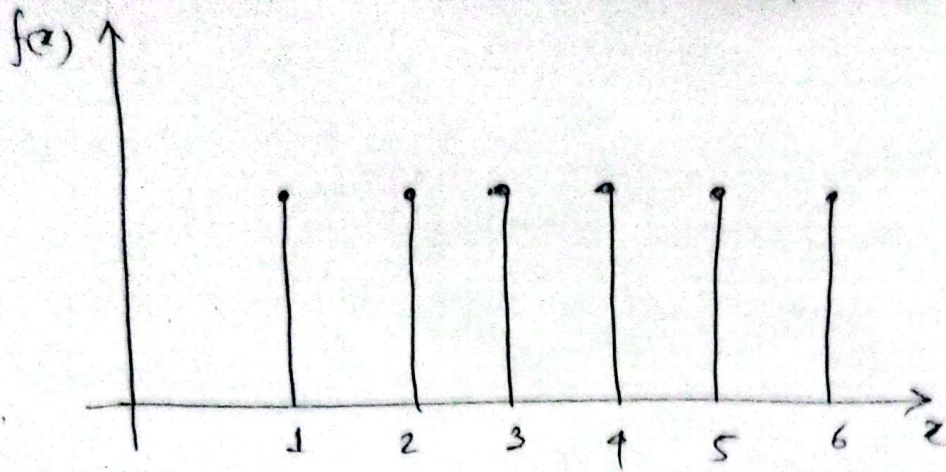
The simplest of all discrete probability distributions is one where the random variable assumes its values with equal probabilities. Such a probability distribution is called rectangular or uniform distribution. It is called rectangular distribution since the histogram of this distribution is a set of rectangles of equal heights. It is uniform because the probability function of this distribution is constant or uniform over the entire range of the variable. Thus, if S is a sample space of a discrete random variable X , and $x_i \in S$ for $i=1, 2, 3, \dots, K$; that is $S = \{1, 2, 3, \dots, K\}$, then $P(x_1) = P(x_2) = \dots = P(x_K) = 1/K$ for a uniform distribution.

Defⁿ:

Let the discrete random variable X assumes the values $x_1, x_2, \dots, x_K$, each with equal probability $1/K$. Then the probability distribution of X , given by

$$f(x; K) = \begin{cases} 1/K; & x=1, 2, 3, \dots, K \\ 0; & \text{otherwise} \end{cases}$$

is known as the uniform or rectangular distribution on the integers.



The distribution represents the outcome of an experiment that is often described by saying that one of the integers $1, 2, 3, \dots, K$ is chosen at random. In this context, the phrase "at random" means that each of the K integers is equally likely to be chosen.

It is easy to verify that $f(x; K)$ is a PMF, since $f(x; K) \geq 0$ and

$$\sum_{x=1}^K f(x; K) = \sum_{x=1}^K \frac{1}{K} = \frac{1}{K} + \frac{1}{K} + \frac{1}{K} + \dots + \frac{1}{K} = 1.$$

Problem:

Let X be a random variable representing the number on a ball drawn at random from a box containing 10 balls numbered 1 to 10. Find the probability distribution of X and hence $P(X < 4)$.

Solⁿ:

Clearly X assumes values $1, 2, 3, \dots, 10$, each with equal probability $\frac{1}{10}$. Hence, the distribution is uniform with $K=10$. Thus,

$$f(x; 10) = \frac{1}{10}; \quad x=1, 2, 3, \dots, 10.$$

• and $P(X < 4) = \sum_{x=1}^3 f(x; 10) = \frac{1}{10} + \frac{1}{10} + \frac{1}{10} = \frac{3}{10}$.

Ans

Problem:

• A fair cubical dice is tossed once. Find the probability distribution of the number of points on the upper face when the dice comes to rest.

Solⁿ:

The sample space for the experiment is $S = \{1, 2, 3, 4, 5, 6\}$. Each element of S occurs with equal probability $\frac{1}{6}$ and hence with $K=6$, the distribution is uniform and thus,

$$f(x; 6) = \frac{1}{6}; \quad x=1, 2, 3, 4, 5, 6.$$

Continuous uniform distribution :

Uniform or rectangular distribution is one of the simplest distributions among the several parametric families of continuous distributions. It is particularly useful in theoretical statistics because it is easily amenable to further mathematical manipulation.

Defⁿ :

When the random variable X is continuous over the interval $[\alpha, \beta]$, then the random variable is said to have the uniform distribution if its density function assumes the form

$$f(x; \alpha, \beta) = \frac{1}{\beta - \alpha} ; \quad \alpha \leq x \leq \beta$$

where α and β are the two parameters of the distribution satisfying the condition $-\infty < \alpha < \beta < \infty$

The particular form of the distribution when X is continuous is often referred to as the uniform - distribution on the interval. Here X represents the outcome of an experiment that is often described by saying that a point is chosen at random from the interval (β, α) .

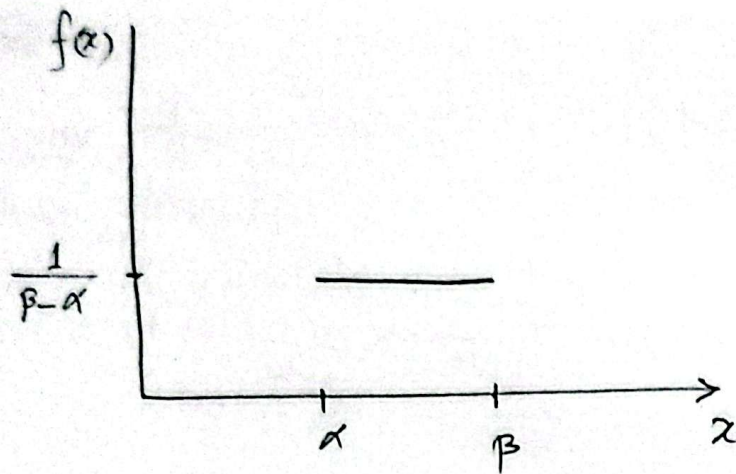


Fig. : Uniform density function

Clearly, $f(x; \alpha, \beta)$ is a PDF since $f(x; \alpha, \beta) \geq 0$ and

$$\int_{-\infty}^{\infty} f(x; \alpha, \beta) dx = \int_{\alpha}^{\beta} \frac{1}{\beta - \alpha} dx = \frac{\beta - \alpha}{\beta - \alpha} = 1$$

Problem:

If X has a uniform distribution in the interval $(0, 10)$ evaluate (i) $P(X < 4)$ and (ii) $P(X > 7)$.

Solⁿ:

Here, $f(x; \alpha, \beta) = \frac{1}{\beta - \alpha} ; 0 \leq x \leq 10$
 $0 ; \text{otherwise}$

For $\alpha = 0$ & $\beta = 10$,

$$f(x; 0, 10) = \frac{1}{10 - 0} = \frac{1}{10} ; 0 \leq x \leq 10$$

$$= 0 ; \text{otherwise}$$

Hence,

$$(i) P(X < 4) = \int_0^4 \frac{1}{10} dx = \frac{4}{10} \text{ Ans}$$

$$(ii) P(X > 7) = \int_7^{10} \frac{1}{10} dx = \frac{1}{10}(10-7) = \frac{3}{10} \text{ Ans}$$

Problem:

Arrivals of customers at a checkpoint follow uniform distribution. It is known that during a given 30-minutes period, one customer arrived at the counter. What is the probability that the customer arrived during the last 5 minutes of the 30-minute period?

Solⁿ:

We are given that the actual time of arrival follows a uniform distribution over the interval $(0, 30)$

Hence, if X denotes the arrival time, then

$$P(25 < X < 30) = \int_{25}^{30} \frac{1}{30} dx = \frac{1}{30} \cdot (30-25) \\ = \frac{1}{6}$$

Ans

Here,

$$f(x; 0, 30) = \frac{1}{30-0} \\ = \frac{1}{30}$$

Problem :

Find the mean, variance, distribution function and moment generating function of the uniform distribution having the density function

$$f(x; \alpha, \beta) = \frac{1}{\beta - \alpha} ; \alpha \leq x \leq \beta$$
$$= 0 ; \text{otherwise}$$

Solⁿ :

$$(i) \text{ Mean} = E[X] = \int_{-\infty}^{\infty} x f(x; \alpha, \beta) dx$$

$$= \int_{\alpha}^{\beta} \frac{x}{\beta - \alpha} dx$$

$$= \frac{1}{\beta - \alpha} \left[\frac{x^2}{2} \right]_{\alpha}^{\beta}$$

$$= \frac{1}{2(\beta - \alpha)} (\beta^2 - \alpha^2)$$

$$= \frac{(\beta + \alpha)(\cancel{\beta - \alpha})}{2(\cancel{\beta - \alpha})}$$

$$\therefore E[X] = \frac{1}{2}(\beta + \alpha)$$

$$\therefore \text{Mean of uniform distribution} = \frac{1}{2}(\beta + \alpha)$$

$$(ii) \text{ Variance} = V[X] = E[X^2] - \{E[X]\}^2 \quad \text{--- ①}$$

Now,

$$E[X^2] = \int_{\alpha}^{\beta} \frac{1}{\beta - \alpha} x^2 dx = \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} x^2 dx$$

$$= \frac{1}{\beta - \alpha} \left[\frac{x^3}{3} \right]_{\alpha}^{\beta} = \frac{1}{3(\beta - \alpha)} (\beta^3 - \alpha^3)$$

$$= \frac{1}{3(\beta - \alpha)} (\beta - \alpha) (\beta^2 + \alpha\beta + \alpha^2)$$

$$= \frac{1}{3} (\beta^2 + \alpha\beta + \alpha^2)$$

From ①, we can write,

$$V[X] = \frac{1}{3} (\beta^2 + \alpha\beta + \alpha^2) - \frac{1}{4} (\beta + \alpha)^2$$

$$= \frac{1}{3} \beta^2 + \frac{1}{3} \alpha\beta + \frac{1}{3} \alpha^2 - \frac{1}{4} (\beta^2 + 2\alpha\beta + \alpha^2)$$

$$= \frac{1}{3} \beta^2 + \frac{1}{3} \alpha\beta + \frac{1}{3} \alpha^2 - \frac{1}{4} \beta^2 - \frac{1}{2} \alpha\beta - \frac{1}{4} \alpha^2$$

$$= \frac{1}{12} \beta^2 - \frac{1}{6} \alpha\beta + \frac{1}{12} \alpha^2$$

$$= \frac{1}{12} (\beta^2 - 2\alpha\beta + \alpha^2)$$

$$\therefore V[X] = \frac{1}{12} (\beta - \alpha)^2$$

∴ Variance of uniform distribution = $\frac{1}{12}(\beta - \alpha)^2$

(ii) Moment generating function, $M_X(t) = E[e^{tx}]$

$$\Rightarrow M_X(t) = \int_{\alpha}^{\beta} e^{tx} f(x; \alpha, \beta) dx$$

$$= \int_{\alpha}^{\beta} e^{tx} \cdot \frac{1}{\beta - \alpha} dx$$

$$= \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} e^{tx} dx$$

$$= \frac{1}{\beta - \alpha} \left[\frac{e^{tx}}{t} \right]_{\alpha}^{\beta}$$

$$\therefore M_X(t) = \frac{1}{t(\beta - \alpha)} (e^{t\beta} - e^{t\alpha})$$

(iv) Distribution function or probability distribution function or CDF is given by

$$F(x) = \int_{-\infty}^x f(x; \alpha, \beta) dx$$

$$= \int_{-\infty}^x \frac{1}{\beta - \alpha} dx = \int_{\alpha}^x \frac{1}{\beta - \alpha} dx$$

$$= \frac{1}{\beta - \alpha} [x]_{\alpha}^x = \frac{x - \alpha}{\beta - \alpha}$$

Therefore, $F(x)$ can be presented as follows

$$F(x) = \begin{cases} 0; & x \leq \alpha \\ \frac{x-\alpha}{\beta-\alpha}; & \alpha < x < \beta \\ 1; & x \geq \beta \end{cases}$$

Problem:

If X is distributed uniformly over the interval $[0, 1]$, find the density function of X and hence its distribution function $F(x)$.

Solⁿ:

Setting $\alpha = 0$ & $\beta = 1$, we obtain

$$f(x; 0, 1) = \begin{cases} \frac{1}{1-0} = 1; & 0 \leq x \leq 1 \\ 0; & \text{otherwise} \end{cases}$$

$$\begin{aligned} \text{and, } F(x) &= \int_{-\infty}^x f(x; 0, 1) dx \\ &= \int_0^x 1 dx = [x]_0^x = x \end{aligned}$$

$$\therefore F(x) = \begin{cases} 0; & x \leq 0 \\ x; & 0 < x < 1 \\ 1; & x \geq 1 \end{cases}$$

Exponential Distribution:

The exponential distribution is a continuous distribution and is a special case of gamma distribution. The exponential distribution has many applications in the field of statistics, particularly in the areas of reliability theory and waiting time or queuing problems.

Defⁿ:

A random variable X is said to have an exponential distribution with parameter α ($\alpha > 0$) if X has a continuous distribution for which the probability density function $f(x; \alpha)$ is specified as follows:

$$f(x; \alpha) = \begin{cases} \alpha e^{-\alpha x} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$$

An alternative form of the distribution is

$$f(x; \lambda) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$$

Problem:

Find the mean, variance and distribution function of the exponential distribution having the density function $f(x; \alpha) = \begin{cases} \alpha e^{-\alpha x} ; x > 0 \\ 0 ; \text{otherwise} \end{cases}$

Solⁿ:

i) The mean of the distribution is

$$E[X] = \int_0^{\infty} x f(x; \alpha) dx$$

$$= \int_0^{\infty} x \alpha e^{-\alpha x} dx$$

$$= - \int_0^{\infty} x d(e^{-\alpha x})$$

$$= - \left\{ \left[x e^{-\alpha x} \right]_0^{\infty} - \int_0^{\infty} e^{-\alpha x} dx \right\}$$

$$= - [0 - 0] + \frac{1}{-\alpha} \left[e^{-\alpha x} \right]_0^{\infty}$$

$$= -\frac{1}{\alpha} [e^{-\infty} - e^0]$$

$$= -\frac{1}{\alpha} [0 - 1]$$

$$\therefore E[X] = \frac{1}{\alpha}$$

ii) Variance of the distribution is

$$V[X] = E[X^2] - \{E[X]\}^2 \quad \text{--- (1)}$$

Now,

$$E[X^2] = \int_0^{\infty} x^2 f(x; \alpha) dx$$

$$\Rightarrow E[X^2] = \int_0^{\infty} x^2 \alpha e^{-\alpha x} dx$$

$$= - \int_0^{\infty} x^2 d(e^{-\alpha x})$$

$$= - \left\{ \left[x^2 e^{-\alpha x} \right]_0^{\infty} - \int_0^{\infty} e^{-\alpha x} \cdot 2x \cdot dx \right\}$$

$$= -[0-0] + 2 \int_0^{\infty} e^{-\alpha x} x dx$$

$$= -\frac{2}{\alpha} \int_0^{\infty} x d(e^{-\alpha x})$$

$$= -\frac{2}{\alpha} \left\{ \left[x e^{-\alpha x} \right]_0^{\infty} - \int_0^{\infty} e^{-\alpha x} dx \right\}$$

$$= -\frac{2}{\alpha} [0-0] + \frac{2}{\alpha} \int_0^{\infty} e^{-\alpha x} dx$$

$$= \frac{2}{\alpha} \left[\frac{e^{-\alpha x}}{-\alpha} \right]_0^{\infty} = -\frac{2}{\alpha^2} \left[e^{-\alpha x} \right]_0^{\infty}$$

$$= -\frac{2}{\alpha^2} [0-1] = \frac{2}{\alpha^2}$$

From ① we obtain

$$V[X] = \frac{2}{\alpha^2} - \left(\frac{1}{\alpha}\right)^2 = \frac{2}{\alpha^2} - \frac{1}{\alpha^2} = \frac{1}{\alpha^2}$$

It can be shown that for the alternative function, $f(x; \lambda)$ the mean is λ and the variance is λ^2 .

Remark:

It is to be noted that the mean and standard deviation of the exponential distribution are equal.

We noted earlier that for the Poisson distribution, the mean and the variance were equal.

iii) The distribution function of the exponential distribution is

$$F(x) = P(X \leq x) = \int_0^x \alpha e^{-\alpha x} dx$$
$$= \frac{\alpha}{-\alpha} \left[e^{-\alpha x} \right]_0^x$$

$$= - \left[e^{-\alpha x} - e^0 \right]$$

$$= 1 - e^{-\alpha x}, \quad x > 0$$

Therefore,

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & ; x > 0 \\ 0 & ; x \leq 0 \end{cases}$$

Example:

The time t seconds between the arrivals of successive vehicles at a particular road crossing has a probability density function

$$f(t) = 0.025 e^{-0.025t}, \quad t \geq 0$$

A person, who takes 20 seconds to cross the road, sets off as one vehicle passes. Find the probability that the person will complete the crossing before the next vehicle arrives.

Solⁿ: For the exponential distribution

$$f(t) = \mu e^{-\mu t} \quad ; \quad P(T > t) = \mu e^{-\mu t}$$

We are given that

$$f(t) = 0.025 e^{-0.025t} \quad ; \quad t \geq 0$$

This is an exponential distribution with $\mu = 0.025$

The person will complete the crossing before the next vehicle arrives if

$$P(T > 20) = e^{-0.025 \times 20} = 0.6065$$

Therefore, the probability that the person completes the crossing before the next vehicle arrives is 0.607.

Th^m:

Suppose X has an exponential probability density function. Show that, if $m, n > 0$, then

$$P[X > (m+n) | X > m] = P(X > n).$$

Proof:

Using the defⁿ of conditional probability of an event, we have

$$P[X > (m+n) | X > m] = \frac{P[X > m+n]}{P(X > m)}$$

because the intersection of the events $(X > m+n)$ and $(X > m)$ is the event $(X > m+n)$ and consequently,

$$P[X > m+n] = \int_{m+n}^{\infty} \lambda e^{-\lambda x} dx = -[e^{-\lambda x}]_{m+n}^{\infty}$$

$$\Rightarrow P(X > m+n) = e^{-\alpha(m+n)}$$

Similarly,

$$P(X > m) = \int_m^{\infty} \alpha e^{-\alpha x} dx = e^{-\alpha m}$$

Thus,

$$\begin{aligned} P[X > (m+n) \mid X > m] &= \frac{P(X > m+n)}{P(X > m)} \\ &= \frac{e^{-\alpha(m+n)}}{e^{-\alpha m}} \\ &= e^{-\alpha n} = P(X > n) \end{aligned}$$

$$\therefore P[X > (m+n) \mid X > m] = P(X > n)$$

This property of the exponential distribution is often known as the memoryless property of the distribution.